SEQUENCE 6

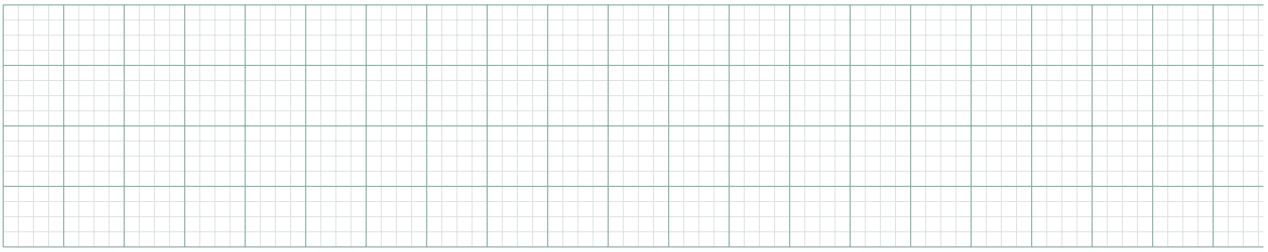
# Functions: representations and reference functions

23 November–4 December 2026

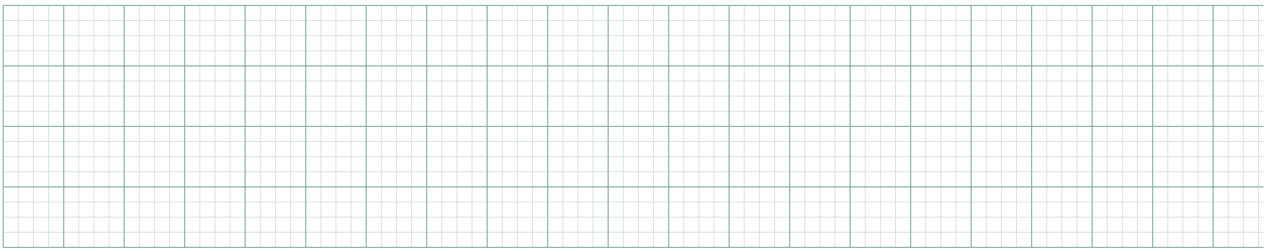
## 6.1 Function, image and preimage

**Key point · Definition.** A real-valued function associates to each real number $x$ of its domain $D_f$ a unique real number $f(x)$, called the image of $x$. A preimage of $y$ is a real number $x \in D_f$ such that $f(x) = y$. We write $f : x \mapsto f(x)$. A number may have no preimage, one, or several.

**Example 1.** Let $g(x) = x^2 - 4x + 32$ on $\mathbb{R}$. Calculate $g(-5)$. Let $h(x) = 3x + 5$; determine the preimage of 6 under $h$.

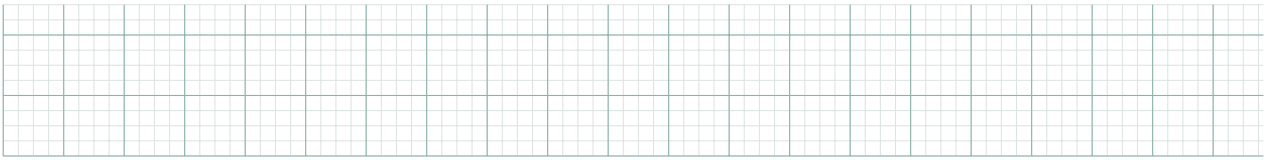


**Example 2.** For the square function, calculate the images of $-3$ and 3, then the preimages of 9, 0 and $-1$.



```
def g(x):
    return x*x - 4*x + 32
```

**Example 3.** Distinguish `g(-5)` from `x = -5`.

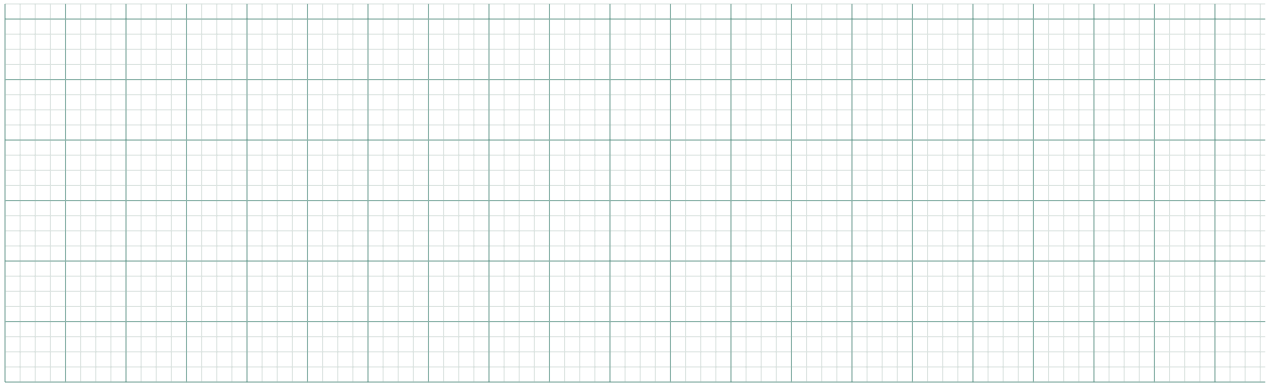


## 6.2 Domain of definition

**Key point · Conditions for existence.** $D_f$ contains the values for which the process defining $f$ makes sense. A denominator must be nonzero; a number under a square root must be positive or zero. Also respect any domain imposed by the statement or the context.

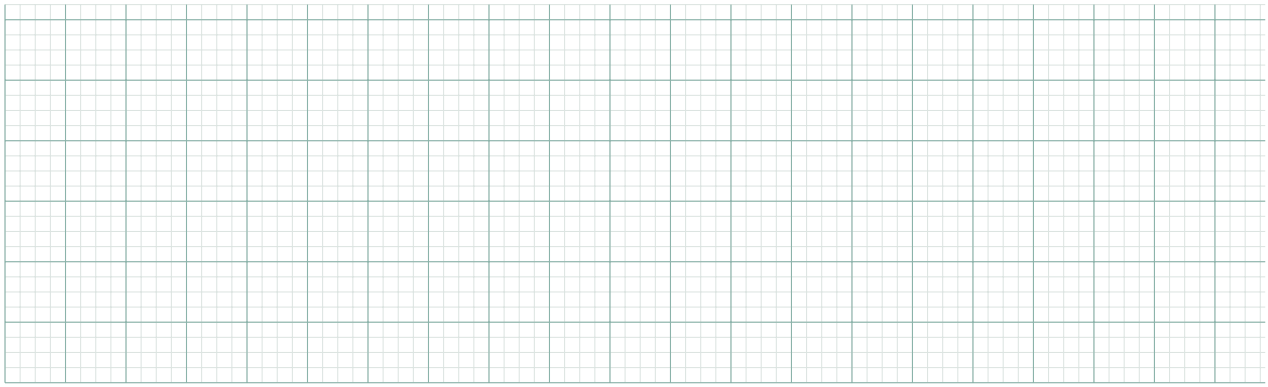
**Example 4.** Determine the domains of definition:

$$f(x) = 3x^2 - 5x + 2, \quad g(x) = \frac{2x+1}{x-4}, \quad h(x) = \frac{x}{x^2-9}, \quad k(x) = \sqrt{2x-6}.$$



**Watch out.** The square root of a negative number does not exist in $\mathbb{R}$; it is not "a negative square root" that is excluded.

**Example 5 · Modelling.** A rectangle of perimeter 20 m has width $x$ m. Express its area and state the domain if both dimensions must be strictly positive.

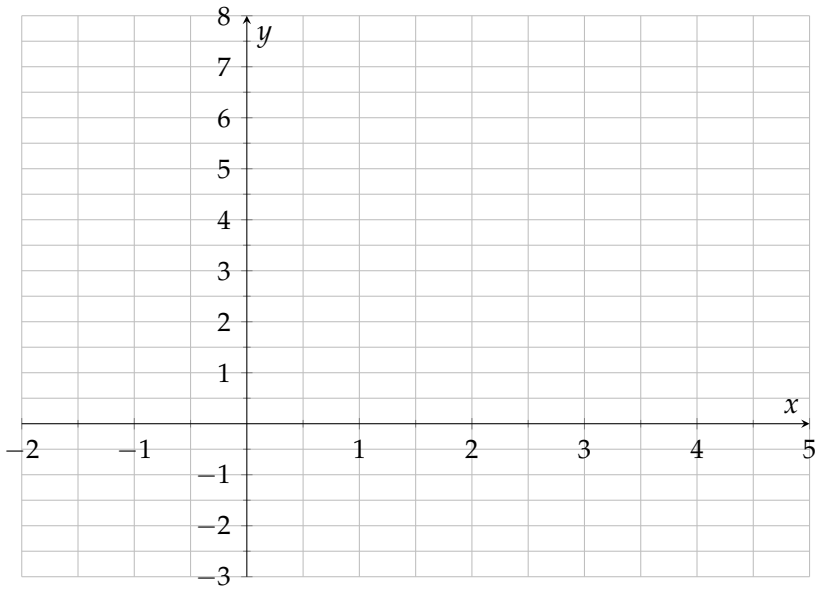


## 6.3 Table of values and representative curve

**Key point · Definition.** $\mathcal{C}_f$ is the set of points $(x, f(x))$, with $x \in D_f$. Thus $M(a,b) \in \mathcal{C}_f \iff a \in D_f$ and $b = f(a)$.

**Example 6.** On $[-1,4]$, $f(x) = x^2 - 2x - 1$. Complete the table, then draw the curve over the whole interval.

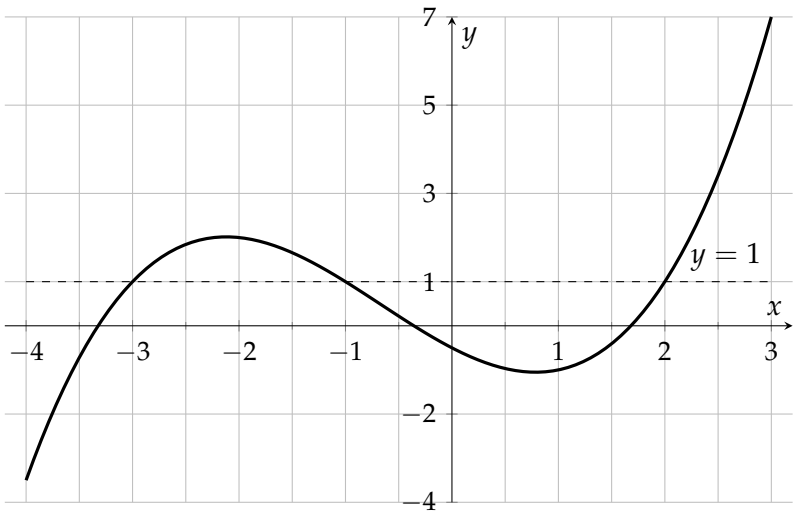
| $x$ | $-1$ | $-0.5$ | $0$ | $1$ | $1.5$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|---|
| $f(x)$ | ..... | ..... | $-1$ | $-2$ | ..... | $-1$ | ..... | ..... |



## 6.4 Graphical solving of equations and inequalities

**Key point · Method.** The solutions of $f(x) = k$ are the abscissae of the intersections of $\mathcal{C}_f$ with $y = k$. For $f(x) \leq k$, read the abscissae where the curve is below this line, intersections included; with $<$, the intersections are excluded.

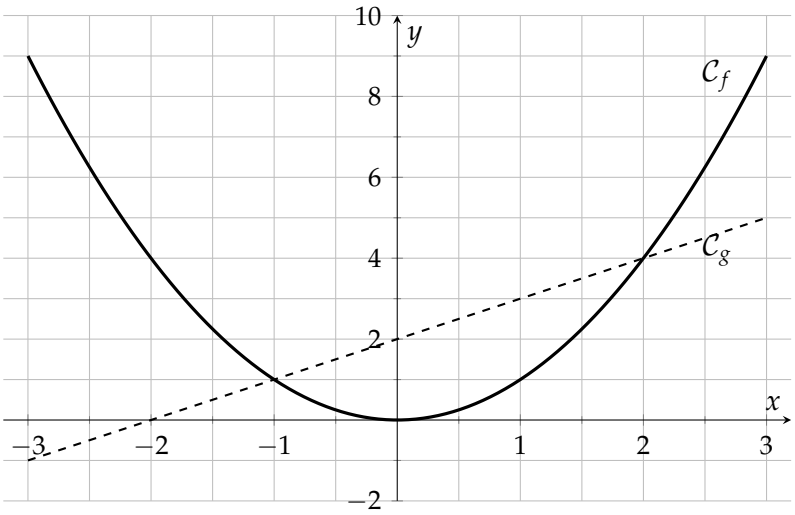
**Example 7.** The curve represents $f(x) = 1 + \frac{1}{4}(x+3)(x+1)(x-2)$ on $[-4, 3]$. Solve graphically $f(x) = 1$, $f(x) \leq 1$ and $f(x) > 1$.



## 6.5 Comparing two functions; reading a sign

**Key point · Method.** $f(x) = g(x)$ is read at the intersections of the curves. For $f(x) < g(x)$, $\mathcal{C}_f$ is strictly below $\mathcal{C}_g$ at the same abscissa. The sign of $f(x)$ is read relative to the x-axis.

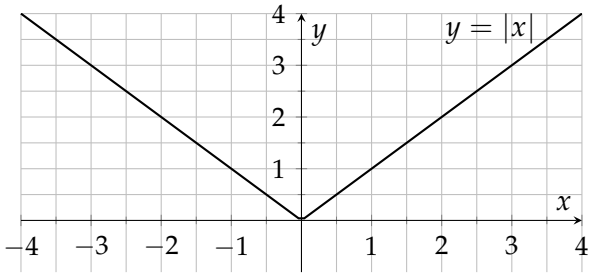
**Example 8.** On $[-3,3]$, $f(x) = x^2$ and $g(x) = x + 2$. Solve $f = g$, $f < g$, $f \geq g$ and determine the sign of $g$.



## 6.6 Affine functions and absolute value

**Key point · Definitions.** An affine function is $f(x) = mx + p$ on $\mathbb{R}$. It is linear if $p = 0$, constant if $m = 0$. For $m \neq 0$, it vanishes at $-p/m$; its sign is that of $m$ to the right of this number. The function $x \mapsto |x|$ is defined on $\mathbb{R}$, positive or zero, and vanishes only at $0$.

**Example 9.** Classify (several answers possible) as affine, linear, constant: $5x + 2$; $5x$; $7 + 2x - 7$; $6$; $6(4x - 2)$; $6x + 5 - 6x$.




**Key point · Equations and inequalities.** If $k < 0$, $|x| = k$ and $|x| \leq k$ have no solution, and $|x| \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

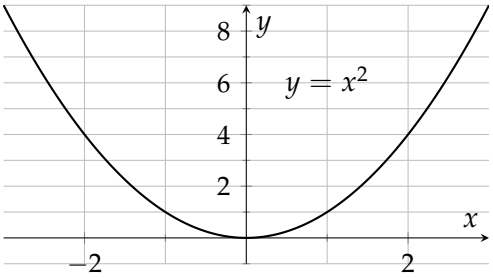
$$|x| = k \iff x \in \{-k, k\}, \quad |x| \leq k \iff x \in [-k, k],$$

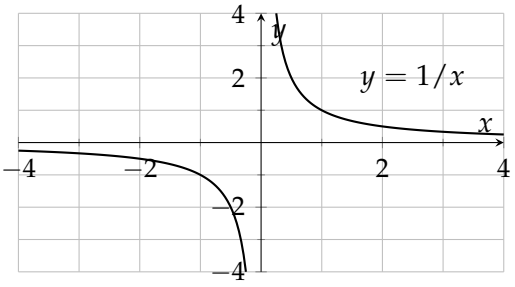
$$|x| \geq k \iff x \in (-\infty, -k] \cup [k, +\infty).$$

**Example 10.** Solve $|x| = 5$, $|x| = -3$, $|x| \leq 3$, $|x| \geq 4$.

## 6.7 Square and inverse functions

**Key point · Definitions and sign.** $x \mapsto x^2$ is defined on $\mathbb{R}$; its curve is a parabola. $x^2 \geq 0$, with equality only at 0. $x \mapsto 1/x$ is defined on $\mathbb{R}^*$; its curve is a hyperbola. $1/x$ has the sign of $x$ and never vanishes.




**Key point · Equations and inequalities with $x^2$.** If $k < 0$, $x^2 = k$ and $x^2 \leq k$ have no solution; $x^2 \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$x^2 = k \iff x \in \{-\sqrt{k}, \sqrt{k}\}, \quad x^2 \leq k \iff x \in [-\sqrt{k}, \sqrt{k}],$$

$$x^2 \geq k \iff x \in (-\infty, -\sqrt{k}] \cup [\sqrt{k}, +\infty).$$

**Example 11.** Solve $x^2 = 4$, $x^2 = -1$, $x^2 \geq 4$, $x^2 \geq -1$, $1/x = 4$, $1/x = 0$.

## 6.8 Inequalities with the inverse function

**Method.** Respect $x \neq 0$ and read both branches of the hyperbola. Do not multiply an inequality by $x$ without knowing its sign.

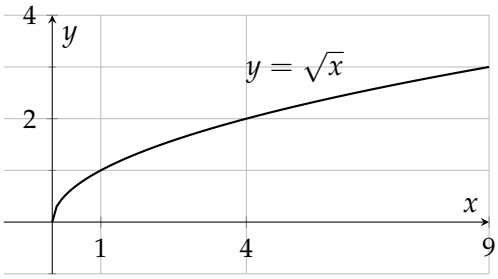
| | $k < 0$ | $k = 0$ | $k > 0$ |
|---|---|---|---|
| $1/x \leq k$ | $[1/k, 0)$ | $(-\infty, 0)$ | $(-\infty, 0) \cup [1/k, +\infty)$ |
| $1/x \geq k$ | $(-\infty, 1/k] \cup (0, +\infty)$ | $(0, +\infty)$ | $(0, 1/k]$ |

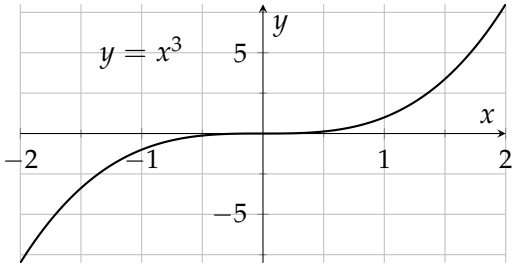
**Example 12.** Solve $1/x \geq 2$, $1/x \leq -1$, $1/x < 1/2$ and $1/x > -2$.

**Example 13.** A student writes: "$1/x \leq 2 \iff 1 \leq 2x$, so $x \geq 1/2$". What is missing?

## 6.9 Using square root and cube

**Key point · Definitions.** $x \mapsto \sqrt{x}$ is defined on $[0, +\infty)$ and takes values that are positive or zero. For $k \geq 0$, $\sqrt{x} = k \iff x = k^2$; for $k < 0$, no solution. $x \mapsto x^3$ is defined on $\mathbb{R}$. Every real number $k$ has a unique preimage, written $\sqrt[3]{k}$.




**Example 14.** Solve $\sqrt{x} = 3$; $\sqrt{x} = -2$; $\sqrt{x} \leq 2$; $\sqrt{x} > 3$; $x^3 = -8$; $x^3 \leq 27$.

**Example 15.** Compare $x^2$ and $x$ for $x = 1/2$, then $x = 2$. Is the square of a positive number always greater than the number itself?